VR-Sets.

An Operational Set Theory Unifying ZFC and ZFA (Brouwer Edition)

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Abstract

We reconstruct set theory inside the VR programme as an operational universe: a set is a *functionality* that, on a query, reveals its members — formally, a pointed graph — and the identity of sets is a *witnessed bisimulation*, carried as a relation and never collapsed to a quotient. On this carrier we obtain, machine-checked in Lean 4 and almost entirely below the `Classical.choice` floor: membership, strong extensionality (identity is exactly co-membership), the empty set, pairing, union, the von Neumann naturals and ω, and the Separation and Replacement schemas. Well-foundedness is not an axiom but a predicate `IsGrounded`: the ZFC universe and the anti-foundational (Aczel) universe are two fragments of one operational domain, not a foundational choice one is forced to make. Version 1.0.3 corrected the central error of earlier versions: the earlier text claimed the operational universe is countable and inferred that the Axiom of Choice is a free theorem and that paradoxes such as Banach–Tarski cannot arise — a Bishop reading that contradicts the operational continuum the cycle built on the Brouwer path. We replace it: the operational universe — sets as becoming — is genuinely non-enumerable (a machine-checked, choice-free diagonal), while the describable register (sets given by a finite rule) is countable. The Axiom of Choice is therefore not an operational theorem; only dependent choice is. The absence of Banach–Tarski is re-explained through conservativity, not countability.

Keywords: operational set theory, non-well-founded sets, bisimulation, strong extensionality, Brouwer, choice sequences, axiom of choice, constructive foundations, Lean 4.

Version note (1.0.4). Editorial synchronisation, no mathematics altered: (i) Remark 5.2 is corrected — VR-Numbers' $\mathbb{R}_{VR}$ is the *computable, countable* reals (its Lean isomorphism with the full classical line is an inexpressibility artifact, VR-Numbers §VIII.6), the genuinely non-enumerable object is the operational becoming-continuum (and this universe, §6.1), and classical $\mathbb{R}$ is a third thing; (ii) references carry correct, separated DOIs (the formal-system *paper* is DOI 10.5281/zenodo.20633314, concept 10.5281/zenodo.20212091; 20324240 is its Lean companion); (iii) the describable register is noted to be a minimal representative of "finite rule" (§6.2); (iv) the single mathlib borrowing is flagged already in §1.

Contents

1 Introduction and the change since Version 1.0.2

VR-Sets realises the set-theoretic layer of the VR programme. A set is not a "thing that contains elements"; it is an *operation* — a functionality that, when queried, reveals members, which are themselves functionalities. This reading is constitutive, not decorative: every notion below (membership, identity, the operations, well-foundedness) is defined in its terms, and every claim is checked by the Lean 4 kernel with its axiom cost recorded.

Versions through 1.0.2 presented the operational universe through a classical surrogate (mathlib's well-founded `ZFSet`) and argued, in their §VI, that the universe is countable, whence the Axiom of Choice holds as a theorem and the classical "pathologies" (non-measurable sets, Banach–Tarski) do not arise. Two things were wrong. First, countability is false of the surrogate (`ZFSet` is the full classical uncountable universe), so the claim was carried by prose, not by the machine. Second, and more deeply, the countability reading is a *Bishop* reading, and it is incompatible with the operational continuum the VR cycle actually constructed, which took the *Brouwer* path: there the space of "becoming" (lawless choice sequences) is non-enumerable, choice-free, and Cantor's diagonal works.

We resolve the tension on the side the cycle already chose. The operational universe of sets is rebuilt from scratch, as functionalities with witnessed-bisimulation identity (§2), and we prove (§6) that this universe — sets as becoming — is not enumerable, by a self-contained diagonal that is choice-free. The countable register is the register of *descriptions*: sets given by a finite rule. The contrast *done countable / becoming non-enumerable* is the corrected §VI. The Axiom of Choice loses its status as a free theorem; this is not a loss but the honest price of a genuinely uncountable operational continuum, and it aligns VR-Sets with VR-Forms (conservativity) and with the operational-continuum work.

What is machine-checked. The development lives in the Lean 4 formalisation of the VR cycle, module `VRCycle.SetsOp` (files `Pointed`, `Builder`, `Closure`, `Omega`, `Extensionality`, `Congruence`, `Becoming`, `Describable`, `Schemas`, `Grounded`). It is self-contained: it imports nothing from mathlib for its core — precisely so that its axiom profile is visible — the *single* mathlib borrowing (the pairing inversion in the describable register) being flagged in Remark 6.5. Throughout, we name the Lean object backing each statement. Almost every object is axiom-free or depends only on `propext`; the single exception is flagged honestly in §6.

2 The carrier: sets as revealing functionalities

2.1 Operational sets

Definition 2.1 (Operational set; `OpSet`). *An operational set is a pointed graph* $x = (V, E, \mathrm{pt})$ *with* $V : \mathrm{Type}$*,* $E : V \to V \to \mathrm{Prop}$*,* $\mathrm{pt} : V$*. Read* $E\,a\,b$ *as "querying the functionality at vertex* b *reveals* a *as a member";* pt *is the set itself. The member of* x *sitting at a vertex* a *is the same graph re-pointed there,* $x.\mathrm{child}\,a := (V, E, a)$*. This is exactly the language in which Aczel's anti-foundation axiom is stated (decoration of a graph), but we take* no quotient*: the identity of sets is supplied operationally, below.*

2.2 Operational identity: witnessed bisimulation, not a quotient

Definition 2.2 (Bisimulation; operational identity $\approx$). *A relation* $R : x.V \to y.V \to \mathrm{Prop}$ *is a bisimulation between* x *and* y *if related vertices reveal matching members in both directions: for all* a, b *with* $R\,a\,b$*,*

$$(\forall a',\ x.E\,a'\,a \Rightarrow \exists b',\ y.E\,b'\,b \wedge R\,a'\,b')\ \wedge\ (\forall b',\ y.E\,b'\,b \Rightarrow \exists a',\ x.E\,a'\,a \wedge R\,a'\,b').$$

We write $x \approx y$ *(*x *and* y *are operationally identical) when some bisimulation relates their points:* $\exists R,\ R\,x.\mathrm{pt}\,y.\mathrm{pt}\ \wedge\ R$ *is a bisimulation.*

Proposition 2.3 ($\approx$ is an equivalence; `Equiv.refl/symm/trans`). $\approx$ *is reflexive, symmetric and transitive. Each is axiom-free in Lean (no* `propext`*, no* `Quot.sound`*, no* `Classical.choice`*).*

The methodological point is decisive. To turn "bisimilar" into a type-level equality — to form the quotient — one must Skolemise the existential matches into functions; that step is exactly VR's transit asymmetry (operational→formal is free, formal→operational has no mechanism), and it is precisely where mathlib's coinductive set construction invokes `Classical.choice`. We therefore do not quotient. Identity remains a relation carried with its witness: a performed bisimulation, not a given completion. This is "doing, not being" made into the construction, and it is why the whole development stays below the choice floor.

2.3 Membership and strong extensionality

Definition 2.4 (Membership; `Mem`). $z \in x$ *iff some member-vertex of* x *is operationally identical to* z*:* $\exists a,\ x.E\,a\,x.\mathrm{pt}\ \wedge\ z \approx x.\mathrm{child}\,a$*.*

Theorem 2.5 (Membership respects identity; `mem_congr`). *If* $y \approx y'$ *then* $z \in y \Rightarrow z \in y'$*. (Axiom-free.)*

Theorem 2.6 (Strong extensionality; `ext`, `equiv_iff_same_mem`). *Two operational sets are identical iff they have the same members:* $x \approx y \iff \forall z,\ (z \in x \leftrightarrow z \in y)$*. Both directions are axiom-free.*

Theorem 2.6 is the heart of the carrier. The forward direction is Theorem 2.5 applied both ways; the backward direction is the coinductive content — one builds a bisimulation out of the member-matching (relate the two roots; relate any further pair whose rooted subtrees are already identical). It is the AFA-style strong extensionality, kept choice-free precisely because identity was never quotiented. It is what licenses $\approx$ as a faithful equality of sets.

3 The operations

All operations are instances of one constructor.

Definition 3.1 (Family supremum; `sup`). *For a family $c : I \to \mathit{OpSet}$, the set $\sup c$ whose members are exactly the $c\,i$ is the pointed graph with a fresh root revealing each component's point, components kept disjoint.*

Theorem 3.2 (`mem_sup`). $z \in \sup c \iff \exists i,\ z \approx c\,i$. *(Axiom-free.)*

From this single law the standard operations follow; each is axiom-free or depends only on `propext`:

- Empty set $\varnothing$ (`emptySup`): the empty family. $z \notin \varnothing$ (`not_mem_emptySup`). $\varnothing$ is the nullary operation — the functionality that reveals nothing — matching the base of the VR formal system.
- Singleton $\{a\}$: $z \in \{a\} \iff z \approx a$.
- Unordered pair $\{a, b\}$: $z \in \{a, b\} \iff z \approx a \lor z \approx b$.
- Union $\bigcup x$: $z \in \bigcup x \iff \exists y,\ y \in x \land z \in y$ (`mem_union`).
- Binary union $a \cup b$ and successor $\operatorname{succ} a := a \cup \{a\}$: $z \in \operatorname{succ} a \iff z \in a \lor z \approx a$ (`mem_succ`).

Proposition 3.3 (Operations respect identity; `*_congr`). *`singleton`, `pair`, $\cup$ and `succ` all respect $\approx$ (through Theorem 2.6).*

3.1 Infinity

Definition 3.4 (Von Neumann naturals and ω; `vn`, `omega`). $\operatorname{vn} 0 := \varnothing$, $\operatorname{vn}(n+1) := \operatorname{succ}(\operatorname{vn} n)$; $\omega := \sup \operatorname{vn}$.

Theorem 3.5 (Infinity; `mem_omega`, `empty_mem_omega`, `omega_succ_closed`). *The members of ω are exactly the $\operatorname{vn} n$ up to $\approx$; $\varnothing \in \omega$; and ω is closed under successor for any member. (Axiom-free, resp. `propext`.)*

3.2 Separation and Replacement

A genuine property or function of sets cannot depend on the particular graph presenting a set; it must respect $\approx$. With that — and no decidability assumption — the two schemas hold.

Theorem 3.6 (Separation; `sep`, `mem_sep`). *For a species p respecting $\approx$, the set $\{z \in x \mid p\,z\}$ exists and $z \in \{z \in x \mid p\,z\} \iff z \in x \land p\,z$. (`propext`.)*

Theorem 3.7 (Replacement; `repl`, `mem_repl`). *For a rule F respecting $\approx$, the image $\{F\,z \mid z \in x\}$ exists and $w \in \{F\,z \mid z \in x\} \iff \exists z,\ z \in x \land w \approx F\,z$. (`propext`.)*

4 Foundation is a predicate, not a choice

Classical set theory must decide, at the level of its axioms, whether $\in$-descent always terminates: ZFC posits Foundation, Aczel's theory posits the Anti-Foundation Axiom (AFA). These are two incompatible commitments, and a classical theory is forced to pick one. VR is not. On the operational carrier, well-foundedness is a property of a set's graph, observed and not postulated.

Definition 4.1 (Groundedness; `IsGrounded`). *x is grounded iff its point is accessible under the reveal relation:* $\mathrm{Acc}\,(E)\,x.\mathrm{pt}$ *— every membership descent from x terminates.*

Theorem 4.2 (Groundedness is a property of the set; `isGrounded_iff`). *If $x \approx y$ then x grounded $\iff$ y grounded. (Axiom-free.)*

Theorem 4.3 (The ZFC fragment is hereditary; `mem_grounded`). *A member of a grounded set is grounded. (Axiom-free.)*

Theorem 4.2 makes "well-founded" a well-defined predicate on operational sets (transported across a bisimulation; the key lemma `acc_bisim` carries accessibility along R). Hence the ZFC fragment is $\{x \mid x \text{ grounded}\}$ ($\varnothing$ is grounded; the fragment is closed downward under $\in$), and the ZFA fragment is the whole universe. The Quine atom $A = \{A\}$ is a perfectly good operational set — one vertex with a self-loop (`quine`), satisfying $A \in A$ (`quine_self_mem`) — and it is not grounded (`quine_not_grounded`); no anti-foundation axiom is invoked, the graph simply has a cycle. VR thus holds ZFC and ZFA natively, as fragments of one universe distinguished by a predicate, with uniform identity (witnessed bisimulation) throughout. The grounded fragment enjoys $\in$-acyclicity: no von Neumann natural is a member of itself (`vn_not_self_mem`, by strong induction).

5 Numbers

The natural numbers are the von Neumann naturals of §3, genuinely distinct as operational sets.

Theorem 5.1 (The operational naturals are distinct; `vn_inj`). $\mathrm{vn}\,n \approx \mathrm{vn}\,m \Rightarrow n = m$. *(`propext`.)*

The proof is the classical one read operationally: the members of $\mathrm{vn}\,m$ are exactly $\mathrm{vn}\,0, \ldots, \mathrm{vn}(m-1)$, no $\mathrm{vn}\,n$ is a member of itself, and these force injectivity by the trichotomy of $\mathbb{N}$.

Remark 5.2 (Three reals — stated honestly). The reals of the VR cycle come in distinct guises that must not be conflated.

- **VR-Numbers'** $\mathbb{R}_{VR}$ is built from Cauchy sequences with a *computable modulus of convergence*; it is isomorphic to the field of *computable* reals and is therefore *countable*. (Its Lean realisation additionally proves an isomorphism with the full classical line, but that is an artifact of Lean's inability to express computability in a single type, not the content of the construction — see VR-Numbers §VIII.6.) This is the *done*/describable side.

- **The operational becoming-continuum** — the space of lawless choice sequences (and the Bishop reals over $\mathbb{Z}$ on the Brouwer path) — is built below the choice floor and is *genuinely non-enumerable*, choice-free. The non-enumerability is real, not classical: it is the same phenomenon this paper proves for the set universe (Theorem 6.1, `universe_not_enumerable`) and that the continuum work proves for $\mathbb{N} \to \mathrm{Bool}$ (`branches_not_enumerable`).

- **The classical line** $\mathbb{R}$ (mathlib) is uncountable in the usual, choice-laden sense.

These coexist. The countable computable reals are the performed/describable register; the non-enumerable becoming-continuum is the operational "$\mathbb{R}$" of the Brouwer path; the classical line is the formal-register referent. Which is canonical for the VR continuum is the programme's open Bishop/Brouwer question; this work commits the set universe to the Brouwer (non-enumerable) reading.

6 Cardinality, the diagonal, and choice — the corrected core

This section replaces the central claims of the earlier §VI.

6.1 The operational universe is non-enumerable

Theorem 6.1 (Non-enumerability of the universe; `universe_not_enumerable`). *No $\mathbb{N}$-indexed enumeration of operational sets is surjective up to operational identity: $\neg\,\exists e : \mathbb{N} \to \mathit{OpSet},\ \forall x,\ \exists n,\ e\,n \approx x$. The proof is choice-free (`propext` only).*

The proof is a self-contained Cantor diagonal inside the operational universe. Given a candidate e, form the diagonal set $D := \{\,\mathrm{vn}\,n \mid \mathrm{vn}\,n \notin e\,n\,\}$, a legitimate operational set (a sup over the subtype of indices n with $\mathrm{vn}\,n \notin e\,n$; no decidability is needed). By the distinctness of the naturals (Theorem 5.1), $\mathrm{vn}\,k \in D \iff \mathrm{vn}\,k \notin e\,k$. If e were surjective, some $e\,m \approx D$, whence by extensionality $\mathrm{vn}\,m \in e\,m \iff \mathrm{vn}\,m \notin e\,m$ — a contradiction.

This is the Brouwerian content of VR-Sets. The diagonal does not fail and the universe is not countable. Its meaning is the cut between the *done* and the *becoming*: the countable register of what has been performed never exhausts the universe of sets as becoming. (In the operational-continuum work the same phenomenon appears for the power set of ω: the characteristic functions $\mathbb{N} \to \mathrm{Bool}$ are non-enumerable, choice-free.)

6.2 The describable register is countable

Definition 6.2 (Finite descriptions; `Desc`, `IsDescribable`). *Let `Desc` be the finite inductive syntax with constructors $\varnothing$, `pair`, `union`, ω, and let `eval`: `Desc` $\to \mathit{OpSet}$ be its interpretation. A set x is describable if $x \approx$ `eval` d for some d : `Desc`.*

Theorem 6.3 (The describable register is countable; `describable_countable`). *There is an $\mathbb{N}$-indexed enumeration reaching every describable set (up to $\approx$), exhibited as $n \mapsto$ `eval`(`decode` n).*

Remark 6.4 (`Desc` is a minimal representative of "finite rule"). `Desc` fixes one concrete finite syntax ($\{\varnothing,$ `pair`, `union`, $\omega\}$, without Separation), so it is deliberately *narrower* than the intuitive "set given by a finite rule". This is exactly why Theorems 6.1 and 6.3 do not collide: the diagonal D of §6.1 is itself given by a finite rule, yet it is *not* expressible in `Desc`. The contrast is robust under this choice: *any* fixed finite syntax yields a countable register, and the same diagonal escapes it. `Desc` is a minimal witness of the schema "done = countable", not an exhaustion of the notion of description.

Remark 6.5 (Honest axiom note). Theorem 6.3 is the only object in the development that depends on `Classical.choice`, and the dependence is *borrowed*, not intrinsic. It enters solely through mathlib's proof that the natural-number pairing is invertible (`Nat.unpair_pair`/`Nat.pair_eq_pair` use choice; `Nat.pair` itself is axiom-free). The fact is constructively true; mathlib merely proves it with incidental choice. We keep the theorem and record the cost. The countability of a register of descriptions is in any case a statement of the *formal register* — meta-accounting over a finite syntax — where classical reasoning is admitted with its cost made visible.

Theorems 6.1 and 6.3 together are the corrected §VI: the earlier "the operational universe is countable, hence AC is free" is false; the correct statement is the contrast — the describable (done) register is countable, the universe (becoming) is not.

6.3 The Axiom of Choice is not a theorem

The earlier §VI argued that AC holds because the universe is countable, so every family can be enumerated and "the first element of each" chosen. That argument is unsound: its premise (every family has a countable enumeration) is exactly what Theorem 6.1 refutes. "AC is free" and "the continuum is genuinely uncountable" are two settings of one switch, and the VR cycle has set it on the Brouwer side. The honest operational picture:

- **Dependent choice** is operational and choice-free: a history-dependent rule determines a branch by recursion (the `operational_dependent_choice` of the continuum work).
- **Weak continuity (WC-N)** holds operationally for finite-information functionals (machine-checked there) and is classically false — the cleanest two-register exhibit.
- **Full AC** — a selection over an uncountable family with no rule — has no operational correlate. It is a label of the formal register, not an operational theorem.

6.4 Why Banach–Tarski does not arise — through conservativity, not counting

The earlier text located the absence of Banach–Tarski in the countability of the universe. With countability gone, the explanation must change — and the correct one is sharper and ties VR-Sets to VR-Forms. The Banach–Tarski construction is a form applied to a form ($T \to T$): in the formal register it builds a completed uncountable totality (the sphere as a set of points) with a selection by full AC (the Vitali set). Neither ingredient has an operational correlate: the completed uncountable totality is a formal-register referent, and full AC a formal-register label. By the invariance of the operational under the application of form — the ground of VR-Forms' conservativity result, itself machine-checked — a formal-register construction does not transit to the operational register: form has no power to manufacture an operational object where none was built. The paradox lives entirely in the formal register and never returns through the operational floor. Its absence is a conservativity phenomenon, not a cardinality accident.

7 Position and honest boundaries

What is established. The operational universe of sets is built without classical scaffolding: a set is a revealing functionality, identity is a witnessed bisimulation (no quotient), and on this carrier membership, strong extensionality, $\varnothing$, pairing, union, ω, Separation and Replacement, the distinctness of the naturals, and the predicate character of well-foundedness are all machine-checked — almost entirely axiom-free or `propext`-only. The corrected cardinality core (universe non-enumerable; describable register countable) is machine-checked, the substantive half choice-free.

What is open or deferred.

- *Bishop vs Brouwer.* The computable $\mathbb{R}_{VR}$ (countable, done) and the non-enumerable becoming-continuum coexist (Remark 5.2); which is canonical for the VR continuum is the programme's open question. This version commits the set universe to the Brouwer reading, consistently with the operational-continuum work.

- *The metalanguage.* Lean treats the underlying types classically; the operational reading is carried in the metalanguage and tracked by axiom profile. "Below the choice floor" is a statement about the construction's audited dependencies, exhibited build-time, not a claim that the ambient logic is intuitionistic.
- *The borrowed choice.* The countability of the describable register inherits one `Classical.choice` from mathlib's pairing inversion (Remark 6.5); removing it would mean re-deriving the arithmetic choice-free, orthogonal to the foundations and left aside.
- *Power set.* There is no operational "power-set operation" returning a completed set; the power set of ω is the becoming, and the non-enumerability of §6 is precisely the operational content of "$\wp(\omega)$". CH is a statement about a formal-register referent; it is read philosophically, not coded.

Relation to the earlier version. Versions through 1.0.2 stand as the classical-surrogate presentation (the ZFC axioms realised over `ZFSet`); their §VI cardinality claims are superseded by §6 here. The mathematics genuinely machine-checked there (the ZFC axioms as closure facts over the surrogate) is unaffected; what is corrected is the operational reading of cardinality and choice.

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- The VR Cycle Lean 4 formalisation, `https://github.com/inventor1975/VRCycle`, module `VRCycle.SetsOp`.